

# Master Plan — Finish Everything, Zero Issues (2026-06-14)

Operator directive (2026-06-14): "what is left unfinished and what are (known) issues we (may) have, prepare comprehensive details and add it in subagents-driven working plan so nothing is left unfinished with zero issues when we finally finish."

This plan is **evidence-backed**, not from memory — every item traces to one of three audits run 2026-06-14 (each a committed, file-cited findings doc):

- docs/audits/2026-06-14-governance-debt-audit.md (173e9747) — constitution/CM-gate debt
- docs/audits/2026-06-14-search-provider-residuals-audit.md (e92590a7) — search/api-app/providers
- docs/audits/2026-06-14-product-known-issues-audit.md (6293ab62) — field/Crashlytics/incidents

**Shipped baseline (done, device-verified):** client 1.3.9-1066 + api-app 0.2.9-14 — the 5-layer search cascade fixed for **no-auth providers** (Internet Archive proven on-device, /v1/{provider}/search → 200, real results). Both §6.AA stages distributed; mirrors at 357f867a. **Shipped builds have ZERO Crashlytics field events** (newest field = 1.3.5/0.2.6); no open crash is on the shipped builds.

**"Definition of Done" (zero issues):** every P0+P1 item below CLOSED with captured anti-bluff proof (device run / falsifiability RED→GREEN / field-confirm), every operator-owed decision made, working tree clean, docs in sync. P2 + infra hardening tracked but not release-blocking.

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## P0 — release-blocking / breaks the operator's actual report

### P0-1 · Auth-provider search (RuTracker / Kinozal) — Auth-Token not attached

**Status:** OPEN, code-confirmed. The operator reported search across **RuTracker + YTS + Kinozal**. The 5-layer fix repaired the Lava-Auth per-instance key (so the request reaches the server), but `ApiBackedTrackerClient.withAuth()` (`core/tracker/client/.../ApiBackedTrackerClient.kt:80-81`) attaches **only** Lava-Auth, never the provider **login session** Auth-Token (`provider:cookie:bb_session=...`, parsed server-side in `lava-api-go/internal/handlers/handlers.go:118`). So `/v1/rutracker/search` arrives anonymous (`Type:"none"`) → RuTracker/Kinozal return login/empty. YTS is curated/no-auth → unaffected. **So search is fixed for no-auth providers only.** **Work:** (a) persist the dynamic provider login session when the user logs in (onboarding/provider-config); (b) thread it as Auth-Token onto `ApiBackedTrackerClient` search/browse/topic/download requests; (c) wire AUTH\_REQUIRED + the login step for `RemoteTrackerDescriptor` (dynamic) providers. **Acceptance:** fresh on-device onboard of RuTracker (+ Kinozal) with **real .env creds** → login OK → search "prince" → the search request carries BOTH Lava-Auth + Auth-Token → HTTP 200 + real rows on screen. **Anti-bluff gate:** falsifiability — drop Auth-Token → auth-provider search goes empty/login; restore → results. A real end-to-end test (holder→factory→Auth-Token on the wire) like `ApiAuthKeyEndToEndWiringTest` for the session header. **Stream:** auth-provider-search (code + tests, then device verify with creds). **Effort:** L. **Blocks:** "search works for the providers reported."

### P0-2 · LVA-008 — search\_input NavBackStackEntry teardown crash

**Status:** OPEN, CRITICAL, gate-RED. Confirmed UPSTREAM androidx-navigation defect; **5 app-level fix classes device-FALSIFIED** (`nav-bump`, `LenientTeardownRule`, nested-host move, `atomic popUpTo`, `NavTeardownGuard`). Real impact: search-then-activity-destroy (rotate / dark-mode / locale / process-death) crashes; gates C06 + C11. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json`. **OPERATOR DECISION OWED (cannot be subagent-resolved):** either (i) file an androidx upstream repro + wait, or (ii) accept-with-§6.AC-telemetry (record the non-fatal, ship) — the 5 in-app fixes are exhausted. **Acceptance:**

operator picks (i)/(ii); if (ii), the telemetry + a tracked-issue citation land + C06/C11 documented as upstream-blocked. **Stream:** operator-decision → then telemetry impl. **Effort:** S (impl) once decided.

### **P0-3 · macOS emulator container/VM path (§6.AH / §6.X-debt) — gates run host-direct**

**Status:** OPEN, BLOCKING (governance audit #1). containerized.go:19–21 hard-requires /dev/kvm; scripts/run-challenge-matrix.sh:215 resolves macOS→host-direct. Every device gate this session ran host-direct (a §6.AH violation — emulators must run in a container/VM). Blocks the §6.AE matrix / §6.Z release canary / tag attestation from being constitution-clean on the operator's darwin/arm64 host. **Work:** add a no-KVM/TCG (or QEMU full-system) runner to submodules/containers/pkg/emulator that boots on macOS; flip run-challenge-matrix.sh/run-release-canary.sh to --runner=containerized|vm with NO host-direct fallback. **Acceptance:** the C00 canary boots + passes via the container/VM runner on macOS; attestation row runner: containers-submodule. **Anti-bluff gate:** a real boot inside the container/VM + a captured per-AVD attestation (not host-direct). **Stream:** containers-submodule (Go, cross-repo pin bump). **Effort:** L.

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## **P1 — high (correctness / operator-requested deliverables)**

### **P1-1 · Watchable issue-videos → ~/Downloads (operator asked repeatedly)**

**Status:** UNDELIVERED. The product audit confirms **none of the 4 issue videos are in ~/Downloads**. issue1/2/4 were re-recorded analyzer-PASS earlier but against 1.3.8 (search broken), so **issue3 (search) was honestly not producible** — search NOW WORKS, so the full set is producible. (Note: an earlier stream reported delivering issue1/2/4; the audit says they're absent — **verify + (re)deliver**, don't assume.) **Work:** re-record paced real-app walkthroughs for issue1 (select-all), issue2 (password eye), issue3 (**search "prince" → real results**, now possible), issue4 (server-list de-dup) on the VM; validate each through HelixQA recording-analyzer (liveness PASS); copy the PASS set to \$HOME/Downloads/ + \$HOME/. **Acceptance:** 4 analyzer-PASS mp4s present in ~/Downloads (verified ls + analyzer verdict committed to evidence). **Anti-bluff gate:** analyzer liveness PASS per video + the on-screen content matches the named flow. **Stream:** video (VM). **Effort:** M.

## P1-2 · YTS / curated search — on-device verification

**Status:** UNVERIFIED on-device (only archiveorg was run). Curated = no-auth, so the transport fix should apply, but unproven. **Acceptance:** on-device onboard YTS → search → real rows (or honest "provider returns nothing for this query" with a different query that has hits). **Stream:** device-verify (folds into P0-1's run). **Effort:** S.

## P1-3 · api-app-uninstalled → silent 401 ("Something went wrong")

**Status:** OPEN. If the api-app is uninstalled, `ApiClient.read()` → null → keyless → silent 401 with only the generic error; no user-visible recovery prompt. **Work:** detect the keyless/uninstalled state on a search failure → surface an actionable prompt ("Start/install the on-device API" or re-onboard). **Acceptance:** device repro (uninstall api-app, search) → actionable UI, not a bare error. **Stream:** client UX + telemetry. **Effort:** M.

## P1-4 · Remote LAN api-app → wrong-key (withLocalApiKeyIfMissing)

**Status:** OPEN, **no covering tests** (`OnboardingViewModel.kt:382`). The helper reads the LOCAL api-app key, which is wrong for a remote/non-loopback api endpoint → 401. **Work:** only read the local key when the endpoint is the local on-device api-app (`host == loopback/own-IP`); for remote endpoints, don't attach the local key. Add tests. **Acceptance:** unit test — remote GoApi endpoint keyless → helper does NOT attach the local key (falsifiability: attach-anyway → wrong-key assertion fails). **Stream:** onboarding code + tests. **Effort:** S.

## P1-5 · RuTracker credential rotation (security, §6.H) — OPERATOR-OWED

**Status:** OWED. Historical git-history leak (nobody85perfect/ironman1985 + RuTor/Kinozal/NNMClub); live surface redacted (d7d4572a), working tree clean — but per §6.H clause 6 the **operator must rotate** the RuTracker (+ siblings) passwords; the git-history purge (§6.T.3 force-push) is also pending operator authorization. (The separate 67th-cycle Firebase-token echo leak is RESOLVED.) **Acceptance:** operator rotates creds + updates `.env`; optional history purge with explicit authorization. **Stream:** operator action (+ I can prep the history-purge plan). **Effort:** operator.

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## P2 — hardening / fidelity / debt (not release-blocking)

| id   | item                                                                                                                                                                                                   | source                        | acceptance                                             | effort   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------|----------|
| P2-1 | <b>L4 READ_API_KEY variant-suffix</b> (\$ {apiKeyPermission} placeholder both apps) so debug+release never collide → test VM exercises the production grant path                                       | residuals #5                  | manifests+build.gradle, clean-install grant test       | S-M      |
| P2-2 | <b>\$6.AC telemetry comprehensive</b> — client key-read failures are logcat-only (not recordWarning→Crashlytics); api-app empty-cursor query() records nothing; Go side has zero non-fatal enforcement | residuals #8, gov \$6.AC-debt | recordNonFatal/recordWarning on the error paths + lint | M        |
| P2-3 | <b>\$6.AB-debt</b> Detekt/go-vet per-feature anti-bluff completeness rule                                                                                                                              | gov                           | rule + hermetic test                                   | M        |
| P2-4 | <b>\$6.Z-debt residual</b> — runtime Gate 6 INSIDE firebase-distribute.sh (SHA/24h/BUILD-SUCCESSFUL), not only pre-push                                                                                | gov                           | gate + tests/ firebase/                                | M        |
| P2-5 | <b>2 FGS Crashlytics issues</b> (9ba8502e+b9baeaede) console close-mark — AFTER 0.2.9 field-confirm                                                                                                    | product #4                    | field-confirm 0.2.9 no-FGS → operator close-marks      | operator |
| P2-6 | <b>\$6.AF chaos/stress (LVA-7)</b> beyond phase-1 + wire into verify-all                                                                                                                               | gov                           | runnable chaos suite                                   | M        |
| P2-7 | <b>Server-side /v1/search SSE aggregator</b> — OPTIONAL (client fan-out sufficient); Go has an unused MultiSearchHandler.GetMultiSearch                                                                | residuals #6                  | decision: implement or delete the dead Go route        | S        |
| P2-8 | <b>\$6.AI</b> thin-index CLAUDE.md restructure + token harness; device-recorder layout; action-prefix LAYER-2 verify                                                                                   | gov                           | per-clause                                             | S-L      |

## Doc-correction (cheap, do first — the docs lie about being behind)

The governance audit found **9 debts marked OPEN in CLAUDE.md that are actually CLOSED** (§6.Y-debt pre-push Check 6 live; §6.AA-debt default flipped firebase-distribute.sh:46; §6.AD scanners exist + verify-all-wired; §6.AE strict-flip check-challenge-coverage.sh:39; CM-COVERAGE-LEDGER strict; LVA-6 codegraph submodules; §6.AI action-prefix hook). **docs/helix-constitution-gates.md is stale (2026-05-16). Work:** correct the stale "open" markers in CLAUDE.md + refresh helix-constitution-gates.md. **Acceptance:** every debt marker in CLAUDE.md matches verified reality. **Effort:** S. **Stream:** docs.

## Working-tree hygiene

git status carries 18 entries: core/apiengine/.../api-source.hash (1-line regen), submodules/tracker\_sdk pointer moved (decide intended-bump vs reset — pins frozen by default), 5 .lava-ci-evidence/stress-chaos/jackett/\*.json (regen churn), untracked raw challenge-video + video-analysis mp4 dirs (curate/gitignore per §11.4.128, never commit raw). **Work:** commit/reset the intended bits, gitignore the raw recordings. **Effort:** S.

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## Operator-owed decisions (CANNOT be subagent-resolved — surface + wait)

1. **LVA-008** (P0-2): file androidx upstream repro OR accept-with-telemetry + distribute.
  2. **RuTracker credential rotation** (P1-5) + optional git-history purge authorization (§6.T.3 force-push).
  3. **FGS Crashlytics console close-marks** (P2-5) after 0.2.9 field-confirm.
  4. **Auth-provider login UX** (P0-1): confirm the intended login surface for dynamic auth providers (onboarding step vs provider-config) before the implementation locks the flow.
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## Execution shape (subagents-driven, parallel where non-conflicting)

- **Wave 1 (parallel, non-conflicting):** doc-correction (docs) || P1-4 remote-key guard+tests (onboarding) || P2-1 L4 permission-suffix (manifests) || working-tree hygiene. All small, disjoint file sets.

- **Wave 2 (P0 code):** P0-1 auth-provider Auth-Token threading (the headline) — code + tests, then a single device-verify run that ALSO covers P1-2 YTS + P1-1 issue3 video (one VM session). P0-3 containers-runner in parallel (cross-repo, Go).
- **Wave 3 (hardening):** P2-2 telemetry || P2-3 Detekt || P2-4 distribute gate || P2-6 chaos.
- **Gate before "finished":** all P0+P1 closed with captured proof; operator decisions made; a final clean build + full device gate (preferably via the P0-3 container runner) + a release if auth-provider search shipped.

**Anti-bluff invariant (every item):** no item is "done" without captured physical evidence — a device run showing the user-visible outcome, OR a falsifiability RED→GREEN with the assertion message, OR a field-confirm. "Compiles / test green against a mock" is never sufficient (this cycle removed 5 such bluffs).